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Driver Interface Research: Describing Driving Performance

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Abstract

Many terms and measures used in automotive research to describe driving performance are inconsistently used and defined, making comparing results across multiple studies difficult. 2015's SAE J2944 outlined functional definitions of driving performance measures and statistics. Now, we are updating to J2944 to reflect currently accepted industry standards and expand the report's scope to include measures for unpaved roadways and off-road scenarios.

Objectives

- Which terms are the most important in automotive engineering?
- How should each of those terms be defined?
- What are some example data, applications, and experiments involving each term?

Methods

- For a given measure of driver performance (e.g.: steering reversal, headway), a literature review was performed.
- The literature review involved the use of Google Scholar to explore publications, references, and citations.
- Findings were organized into a report providing definitions of terms, application guidance, and references.

Example: headway can be defined in many ways

- Failure to clarify specifics of how headway is measured is detrimental and confusing
- Time headway and Time Gap lead to different results, see figure
- In this example 17.4% error if Time Headway (option A) is misrepresented and measured as Time Gap (Table 1)

Table 1. Time Gap & Headway

	Time Gap	Time Headway
Mean	1.278	1.548
SD	0.569	0.732

Table 2. Statistics for Mean Lane Departure (s)

	Assist	No Assist
Option B	4.17	5.63
Option D	4.73	6.64

Discussion

In similar future research, it may be beneficial to develop a software system to automate data collection and synthesis, as opposed manually.

A common set of definitions for describing driving performance will facilitate the comparison of results across different projects. Increasing standardization will result in higher quality research; leading to improvements in safety, efficiency, and the sustainability of motor vehicles.

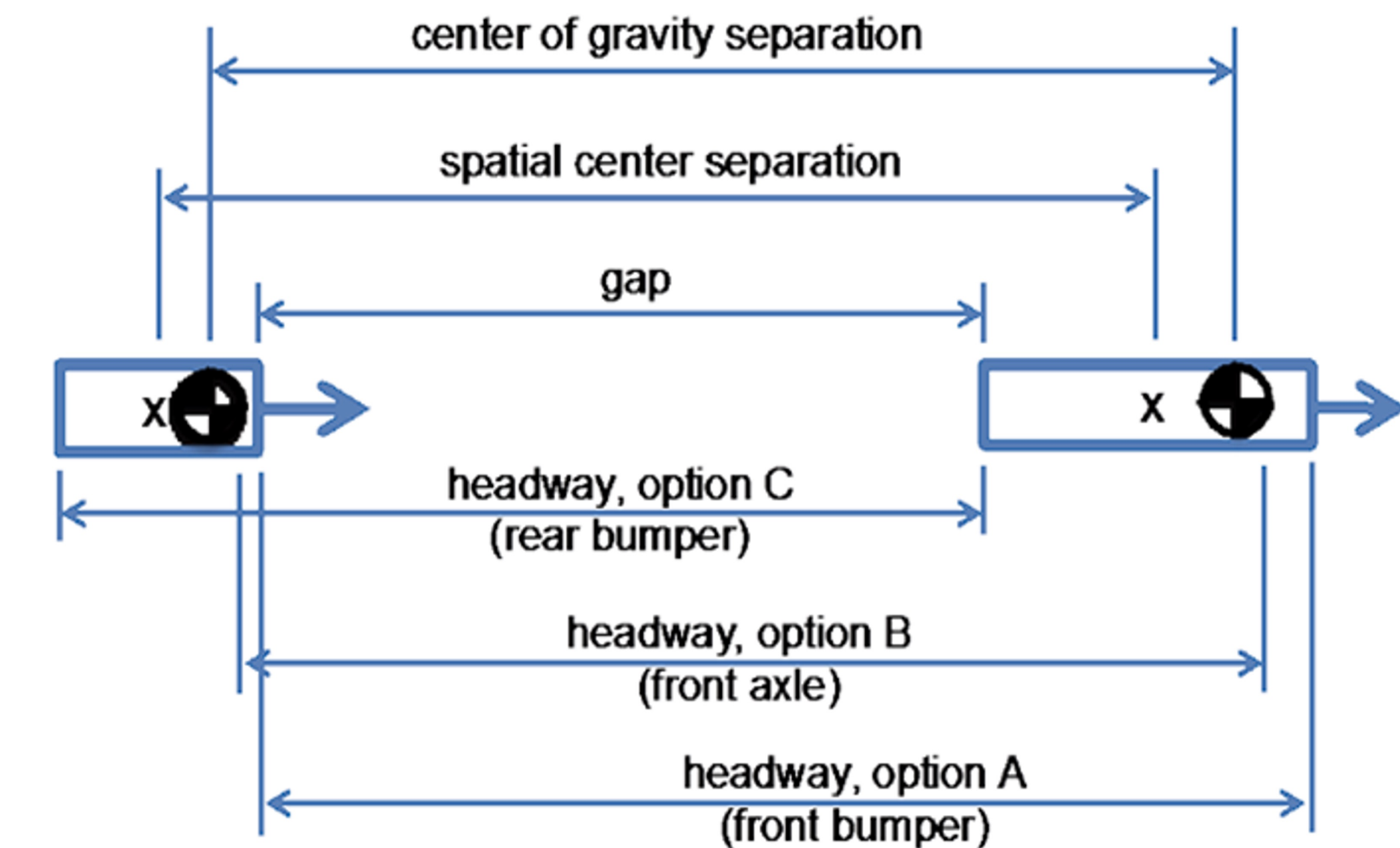


Figure 1. Longitudinal time & distance measures

Table 3. Headway data w/ varying traffic flow

Strategy	W/O OEDVP		W/ OEDVP	
Period	Peak	N-Peak	Peak	N-Peak
Headway mean(s)	3.70	2.82	2.91	3.11
Headway SD (s)	4.21	1.61	1.76	2.10
Min (s)	0.62	0.64	0.35	0.65
Max (s)	60.00	10.18	13.00	17.40

Conclusions

- We identified roughly 150 terms and measures to standardize and define
- For each term, we provide 1-3 definitional alternatives
- For each term, we provide 1-3 reference papers

References

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